

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	[A particle's] maximum displacement [from the equilibrium position] Accept distance from {crest / trough} to {midpoint / rest / equilibrium point} Don't accept maximum height of wave if unqualified	1			1		
		(ii)	$f = 50 \text{ [Hz]}$ or $T = 0.02 \text{ [s]}$ (1) Use of $v = f\lambda$ or $v = \frac{\lambda}{T}$ (1) $= 7.5 \text{ m s}^{-1}$ or 750 cm s^{-1} unit mark (1)	1	1 1		3	3	
		(iii)	Sinusoidal wave with same A and T as original (1) Starting at (0,6) i.e. cos function drawn (1) Minimum of 1 wave drawn		2		2	2	
	(b)		Indicative content: Formation of stationary wave: - two [progressive] waves travelling in opposite directions - same frequency / wavelength / amplitude - reference to pulley / fixed point to act as reflector - interference / superposition - nodes and antinodes Stationary / progressive comparison: - Stationary waves have the same frequency or wavelength as the progressive waves which created them - Particles oscillate in both - ENERGY: not transferred (S) / transferred (P) - PHASE: neighbouring particles [in same loop] in phase (S) / neighbouring particles not in phase / continuous change of phase along the wave (P)	6			6		

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				AO1	AO2	AO3	Total	Maths	Prac
			• WAVELENGTH: $\lambda = 2 \times$ internodal distance (S) / $\lambda =$ distance between 2 consecutive points in phase (P) • AMPLITUDE: Neighbouring particles have different amplitudes or amplitude is zero at nodal points (S) / Amplitude the same for all particles on the wave (P) • FREQUENCY: all particles vibrate at same frequency except at the nodes (S) / all particles vibrate at same frequency (P) 5-6 marks A comprehensive description is provided on both the <i>formation of a stationary wave</i> and <i>comparison</i> given. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks A comprehensive description is provided on either the <i>formation of a stationary wave</i> or <i>comparison</i> or a limited description is provided of both. <i>There is a line of reasoning that is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks A limited description is provided on either the <i>formation of a stationary wave</i> or <i>comparison</i>. <i>A basic line of reasoning that is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks No attempt made or no response worthy of credit.						
			Question 3 total	8	4	0	12	5	0